CIND820 Milestone 3

Julia Wickens

501406008

Supervisor: Dr. Ceni Babaoglu

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Toronto Metropolitan University

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Introduction

Job-Simulation Track

Supply Chain Analyst at a Canadian retail company that imports products from Asia

Problem/Decision Context

Retail companies must receive imported inventory on time to prevent stocking out. Stockouts not only lead to loss of revenue for the company, but also potentially prevent customers from receiving the things they need. When importing inventory from overseas, lead times can be greatly affected by disruptions such as strikes, natural disasters, and other major world events. These delays interrupt the flow of inventory and cause a lack of predictability for retailers and customers.

Research Questions

1. How did the COVID-19 demand surge (August 2020-November 2021), the B.C. floods (December 2021-February 2022), and the ILWU port strike (July-August 2023) impact the Shanghai-to-Toronto shipping transit time when measured against the 12 months preceding the event's onset? How quickly did the average lead times increase, and how many days were added at the peak of the issue?
2. How much of the added transit time was due to port dwell at the West Coast ports versus other stages (ocean and rail)?
3. How many extra days would have needed to be added to the transit lead time during each of these events to cover 95% of months? How much additional lead time should a planner anticipate needing in the first month of a similar future disruption, before its full impact is known?

Description of Dataset

This project uses Transportation Supply Chain Performance (Statistics Canada Table 23-10-0271-01), published on the Government of Canada Open Government Portal. It contains monthly performance indicators for Canada's freight transportation supply chain, covering January 2017 to September 2024. The indicators used in this project are:

- End-to-end container transit time, Shanghai to Toronto via West Coast ports (days)
- Port dwell time, West Coast ports (days)

The full file also includes port dwell time at East Coast ports, grain-transportation indicators, and port-specific commodity flow indicators (Halifax, Montréal, Prince Rupert, Vancouver). These are not used in this project as they are not relevant to a general-merchandise import route. The East

Coast ports do not have an equivalent end-to-end measure that can be used to show the timing of the full transit journey, so only the West Coast ports are used.

Source and Provenance Trail

Source: Transportation Supply Chain Performance, Table 23-10-0271-01, Statistics Canada, published via the Government of Canada Open Government Portal.

Link: https://open.canada.ca/data/en/dataset/59700495-8526-403c-88b6-b0fa8d574297 (table view: https://www150.statcan.gc.ca/t1/tbl1/en/tv.action?pid=2310027101)

Accessed: September 23, 2026

Provenance Trail: Statistics Canada's table description states that the indicators 'draw from a variety of data sources' but does not name them in the publicly visible summary. The data is collected and validated by Transport Canada before transferring the data to Statistics Canada. Statistics Canada publishes these compiled indicators as an official data table (23-10-0271-01), which is listed on the Government of Canada Open Government Portal. I downloaded the table directly as a CSV file from the Statistics Canada table page on September 23, 2026.

Data Preparation

The raw dataset was downloaded in long format, with one row per date-indicator combination across all performance indicators in the table. For this project, the two relevant series, end-to-end transit time (Shanghai to Toronto via West Coast ports) and West Coast port dwell time, were extracted and pivoted into wide format, producing one row per month with each series as its own column. No values were modified in any way during this process. This was done to remove irrelevant information for the data profiling process.

For the coding portion of this project, the “date” column was converted from text to date data type. The additional derived value of “non-dwell days” was also added to the data, created by subtracting the port dwell time from the end-to-end transit time. No scaling, encoding, sampling, or imputation was applied.

This dataset is structured, as it has fixed, consistent columns and each line represents one observation in a uniform structure with no variation in structure from row to row. This dataset has 93 rows, one per month from January 2017 to September 2024. There are no missing values in this dataset. Outliers were not removed as they are important to the analysis being performed in this case.

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Data Analysis

Univariate Analysis

Analysis was conducted using fg-data-profiling in Python. The number of data points for each variable is 93.

Date

Data type	ordinal categorical (year-month)
Min	2017-01
Max	2024-09
Missing	0
Invalid	0

West Coast Dwell

Data type	Continuous numerical (average days by month)
Min	1.9
Max	9.2
Range	7.3
Missing	0
Invalid	0
Mean	4.4236559
Standard Deviation	1.4808486
Variance	2.1929126

Q1	3.4
Q3	5.4

End-to-end Transit Days (Shanghai to Toronto)

Data type	Continuous numerical (average days by month)
Min	24
Max	50
Range	26
Missing	0
Invalid	0
Mean	31.303226
Standard Deviation	5.6072052
Variance	31.44075
Q1	26.3
Q3	34.7

Non-Dwell Days

Data type	Continuous numerical (average days by month)
Min	20.8
Max	42.7
Range	21.9
Missing	0

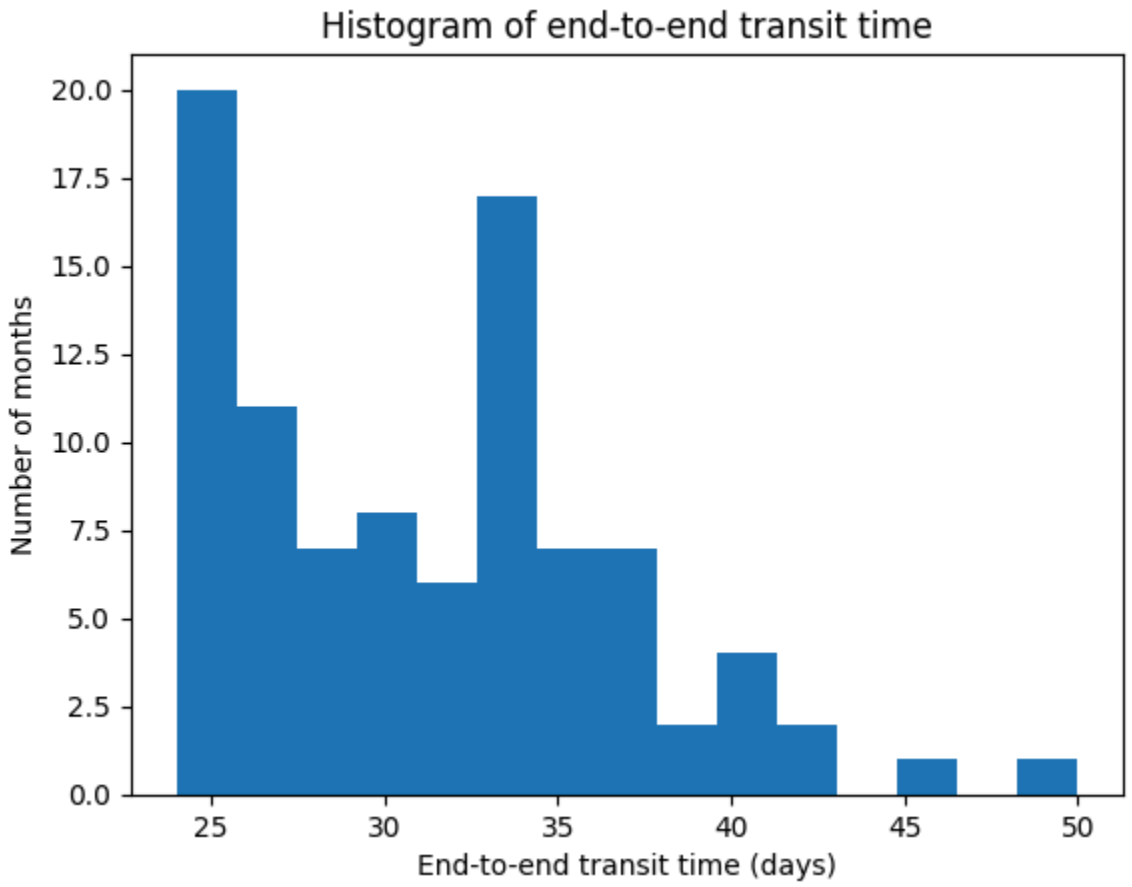
Invalid	0
Mean	26.879570
Standard Deviation	4.637408
Variance	21.505556
Q1	22.6
Q3	30.2

Outliers

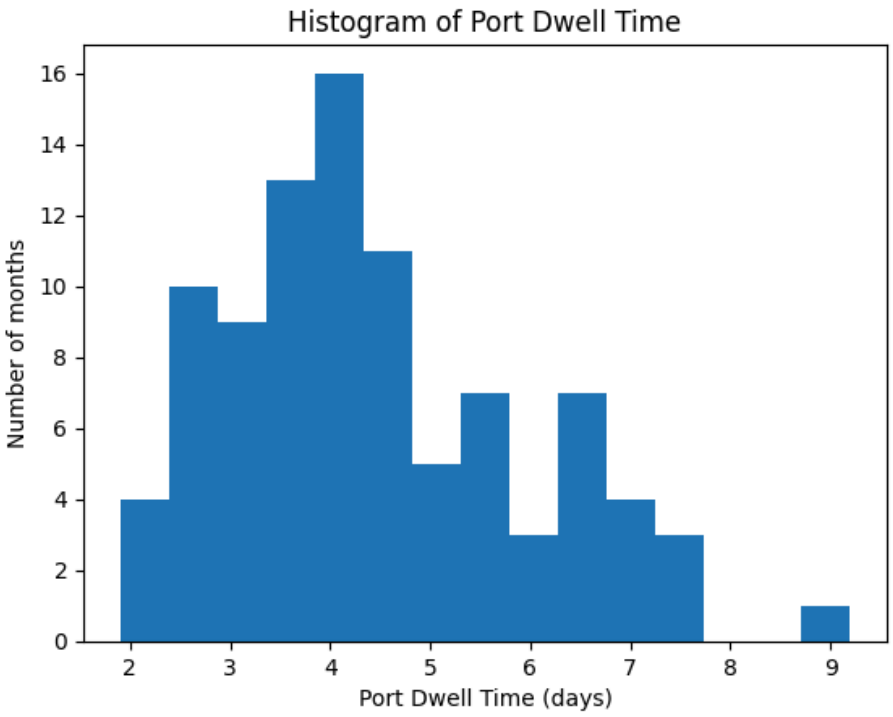
Column	Month	Value
End-to-end Transit Days	2022-01	50
West Coast Port Dwell	2023-01	9.2
Non-Dwell Days	2022-01	42.7

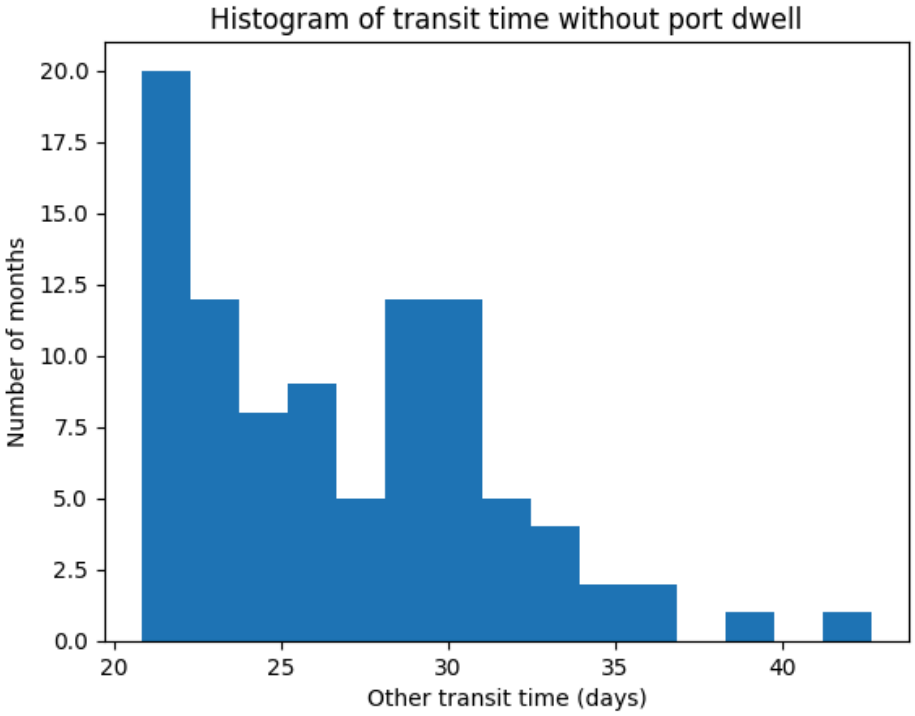
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Histograms



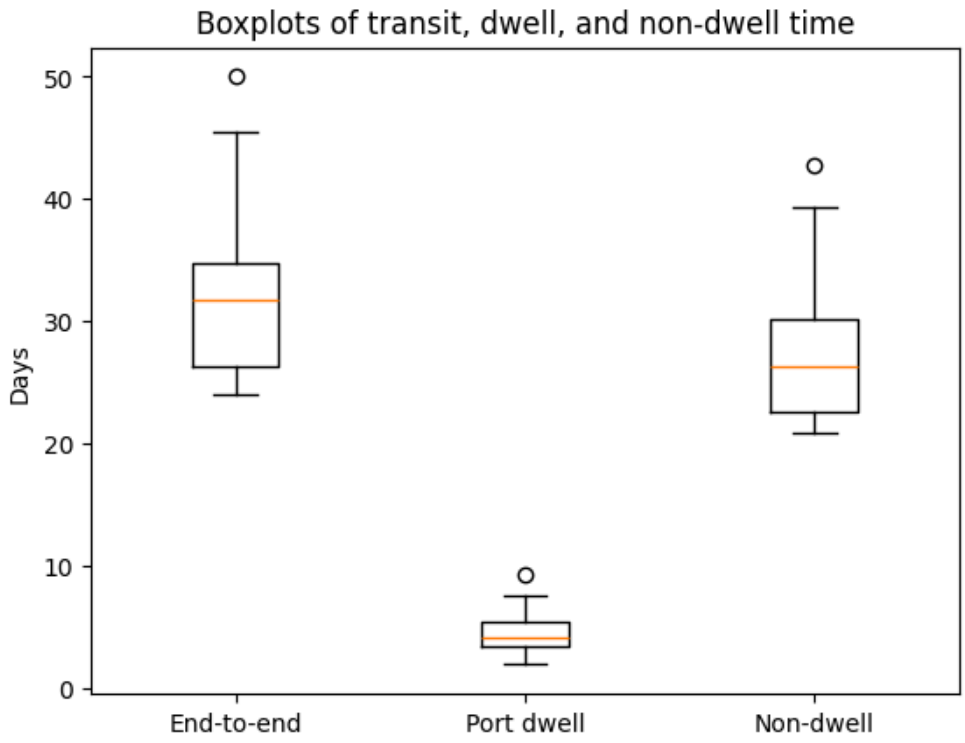
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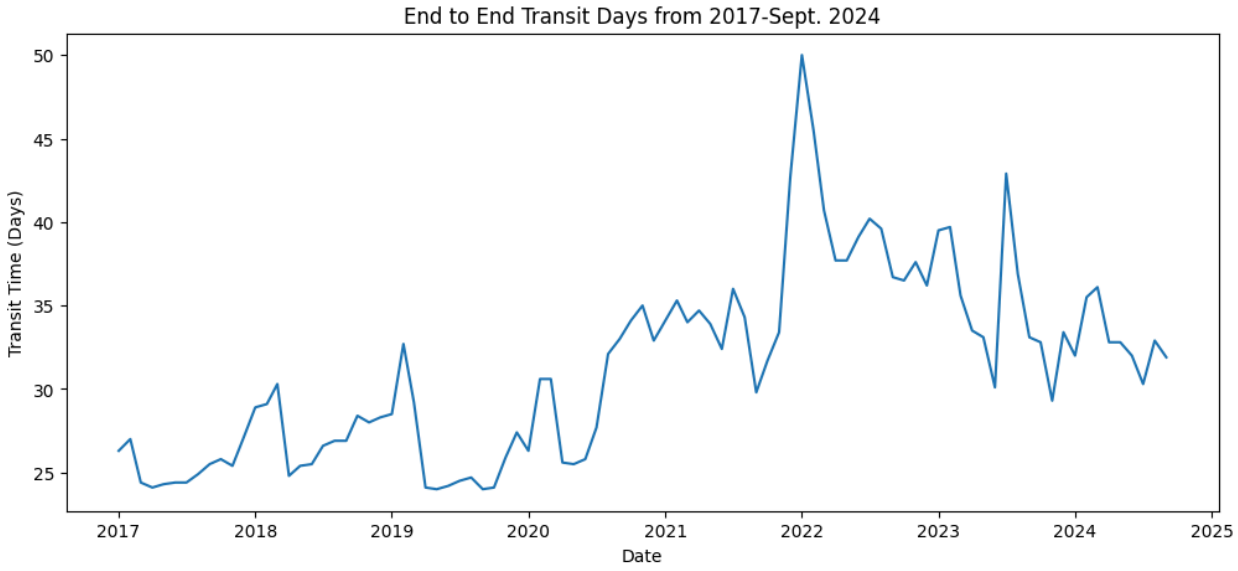


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Boxplots



Chronological Plot of End-to-End Data



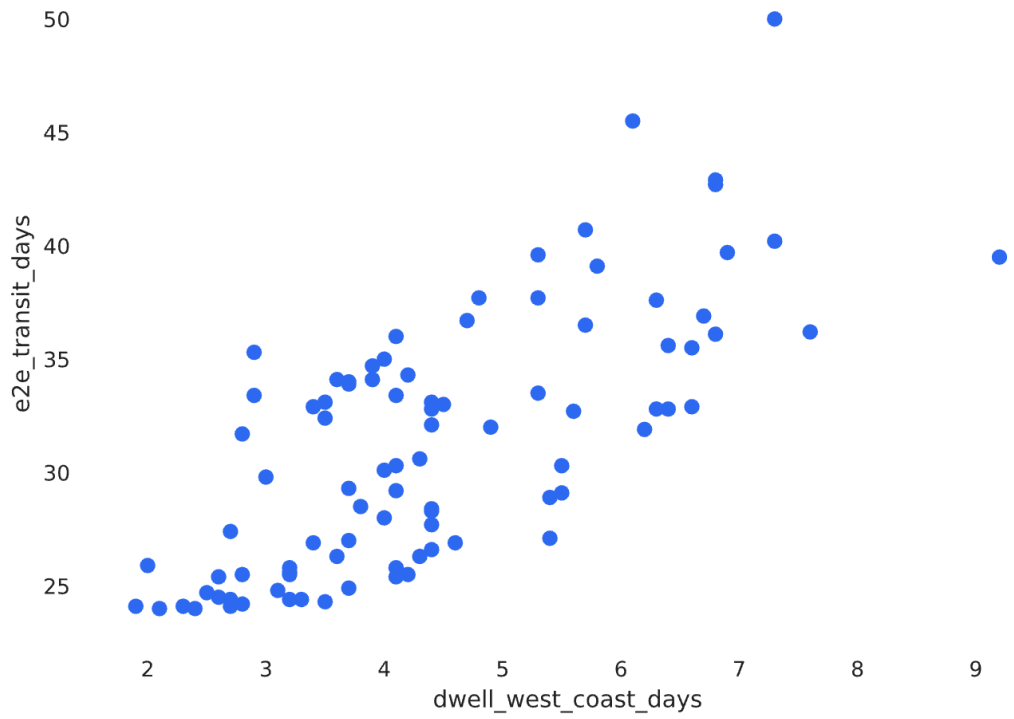
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Bivariate Analysis

Pearson coefficient: `0.7303077332374569`

Spearman coefficient: `0.7172560397968121`

Scatter Plot



Patterns and Implications

1. After the COVID surge, transit lead times never recovered to where they were before 2020. The average end-to-end transit time before the COVID-19 pandemic was 26.5 days. In the years following the pandemic, annual averages ranged from 32.9 to 39.8 days. The implication of this is the need for 12-month average benchmarks for each event being tested rather than a singular benchmark to use for all of the events, as what is considered a “normal” lead time changes over time.
2. There is a noted increase in lead time every January-March prior to the pandemic. This can be attributed to scheduling around Lunar New Year, the effects of winter weather,

and increased demand during peak season, but the data does not separate these factors. This is a noted limitation that is not accounted for in this project.

3. There is one outlier in port dwell time in January 2023 that is not explained by any of the events being studied in this project. This could be the impact of a safety investigation that resulted from a fatal accident at the port during this time, but this has not been confirmed by any sources.
4. Dwell time and end-to-end transit time are highly correlated, but there are some differences. This supports the need for research question 2, which analyzes the respective impact of dwell time and non-dwell time on overall transit time.

Impact Measurement and Decomposition

Event	COVID Surge	BC Floods	Port Strike
Benchmark lead time (end-to-end days)	26.52	33.54	36.53
Peak deviation (days)	9.48	16.46	6.37
Duration of deviation (months)	16	3	2
Months from start to peak	12	2	1
Rate of increase (days per month)	0.79	8.23	6.37
Dwell contribution % to benchmark	11.8%	10.3%	16.5%
Dwell contribution % to deviation	7.4%	26.1%	21.7%
Non-dwell contribution % to benchmark	88.2%	89.7%	83.5%
Non-dwell contribution % to deviation	92.6%	73.9%	78.3%

Lead time buffer required (days)	9.48	16.46	6.37

Note: It was found that the peak month for the COVID surge was late due to the shape of the spike for this event. Unlike the others, it rose and plateaued over a number of months rather than spiking and coming back down. This is noted as it will impact the usability of the peak data for this event.

Analysis

Peak transit time rose by 9.5 days for the COVID surge, 16.5 days for the B.C. floods, and 6.4 days for the ILWU port strike. Port dwell made up a larger share of the added time in the floods (26%) and strike (22%) than in COVID-19 (7%). This is likely because the floods and the strike were local events that impacted the ports in BC, while COVID-19 was a global event that had a greater impact on overall transit time.

The floods had the largest and fastest increase. Transit time reached 16.5 days above benchmark (49%) in January 2022, two months after onset. The strike peaked at 6.4 days (17%) in its first month. COVID peaked at 9.5 days (36%) twelve months in. All three events reached more than half of peak deviation within the first month. As noted above, the COVID rate does not properly reflect the initial increase, as it spiked and then plateaued for several months rather than coming back down.

Because the event windows contain only 16, 3, and 2 months, covering 95% of months means covering all of them, so the buffer equals each event's peak (9.5, 16.5, and 6.4 days).

Baseline Evaluation

This test asks whether the deviation in the first month of an event could have been used to anticipate how large that event's peak would become, using the other events as predictors.

Each of the three events is held out as a test case, and the two remaining events are used to build the rule, which is then used to predict the held-out event's peak deviation based on the deviation in the first month of the held-out event. Only the first month is used because the flood and strike windows are 3 and 2 months long, respectively, and a longer signal would include the peak month in the predictor.

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Three tests were compared:

1. **peak_first:** assumes the peak equals the first month's deviation, meaning the disruption immediately spikes to its peak.
2. **avg_peak:** uses the average peak of the two training events to predict the held-out event and ignores the first month.
3. **scaling:** takes the average ratio of peak to first-month deviation from the two training events (COVID 1.70, floods 1.80, strike 1.00) and multiplies it by the held-out event's first-month deviation.

Each benchmark uses only the 12 months before its own event. A held-out event's data is never used in its own prediction. Events are treated as whole units, not shuffled months, as they are time-ordered data. The metric is absolute error in days, as the output is measured in days, so error metrics are the appropriate family. Accuracy, precision and recall apply to classification and do not fit a continuous target.

Results:

Held-out event	**Actual peak**	**Using first month (predicted/error)**	**Using average of other two events (predicted/error)**	**Scaling (predicted/error)**
COVID surge	9.48	5.58 / 3.90	11.41 / 1.93	7.80 / 1.68
BC floods	16.46	9.16 / 7.30	7.92 / 8.54	12.36 / 4.10
Port strike	6.37	6.37 / 0.00	12.97 / 6.60	11.13 / 4.76

Interpretation

- The scaling rule had the lowest error for both COVID and the BC floods. This suggests the first month carries some information about the eventual peak, but is not the only predictor of peak.
- Using the first month scored 0.00 for the strike only because the strike's peak is its first month, so this result says nothing about the rule.

- The scaling rule predicted too low for COVID (7.80 vs 9.48) and the floods (12.36 vs 16.46) and too high for the strike (11.13 vs 6.37). The events that built up after a first month were underestimated, and the one that spiked at once was overestimated.
- The ratios (1.70, 1.80, 1.00) differ, so event type appears to matter as a predictor.

The test is a first proof of concept. It supports using early deviation as a starting signal, but three events cannot show how reliable the rule is, especially as they are different in terms of type and scale.

Limitations and Next Steps

Limitations

Baseline test

- Only three test cases, so results are individual comparisons and not a validated accuracy figure.
- The scaling rule's errors on the floods (4.1 days) and strike (4.8 days) are large relative to peaks of 6 to 16 days.
- COVID's peak is a single month on a plateau, so its error is less meaningful than the others.
- The strike's peak is its first month, so using the first month as a peak scores 0.00 for it by construction.
- Only the first month is used as the early signal, because the floods and strike windows are 3 and 2 months long.

Data and method

- One route (Shanghai to Toronto via West Coast ports) and three events, so results don't generalize to other routes or disruption types.
- Monthly averages cannot show short disruptions, such as the roughly 4-day rail lockout in August 2024, and the buffers cover average monthly conditions, not individual shipments.
- Winter seasonality (January to March) overlaps the floods window and can't be separated from the events.
- The strike's benchmark (36.5 days) includes the elevated period before it, so its measured impact is probably understated.
- Non-dwell time is a calculated remainder, not a measured ocean and rail time.

Next Steps

1. Test how sensitive the benchmarks and baseline errors are to the benchmark window (e.g. 6 and 18 months).
2. Compare a different early signal, such as percent above benchmark.
3. Review the January 2023 dwell value further.

AI Note

GenAI was used during the design stage of this project for topic and research question idea-generation, and proofreading of this written submission.

It was additionally used during the architecture and data audit stage for formatting citations, proofreading, and troubleshooting while exporting the HTML for the github repository.

It was used during the coding portion for troubleshooting code, as well as the use of prompts to assist with the base code for the baseline evaluation, which was then reviewed, commented, and tested manually for verification.

Verification Plan

1. No final results will rely on AI-generated calculations, they will all be produced by code that I have written and run myself.
2. The only data used will be from the dataset produced by the Canadian Government, none will be fabricated using AI.
3. All facts related to events, dates, and other details were independently verified by me before being included in this report.
4. All decisions including which data to include, the methodology for analysis, and any conclusions from the data have been and will be made by me, without the use of AI.

Links

Github repository: https://github.com/juliawickens3/CIND-820

Video link:
https://drive.google.com/file/d/1laoaENrl_OhaEblyDal6_H_oVGSMaHxL/view?usp=sharing